

Optical Engineering

OpticalEngineering.SPIEDigitalLibrary.org

High-throughput machining using a high-average power ultrashort pulse laser and high-speed polygon scanner

Joerg Schille
Lutz Schneider
André Streek
Sascha Kloetzer
Udo Loeschner

High-throughput machining using a high-average power ultrashort pulse laser and high-speed polygon scanner

Joerg Schille,* Lutz Schneider, André Streek, Sascha Kloetzer, and Udo Loeschner

University of Applied Sciences Mittweida, Laserinstitut Hochschule Mittweida, Technikumplatz 17, 09648 Mittweida, Germany

Abstract. High-throughput ultrashort pulse laser machining is investigated on various industrial grade metals (aluminum, copper, and stainless steel) and Al_2O_3 ceramic at unprecedented processing speeds. This is achieved by using a high-average power picosecond laser in conjunction with a unique, in-house developed polygon mirror-based biaxial scanning system. Therefore, different concepts of polygon scanners are engineered and tested to find the best architecture for high-speed and precision laser beam scanning. In order to identify the optimum conditions for efficient processing when using high-average laser powers, the depths of cavities made in the samples by varying the processing parameter settings are analyzed and, from the results obtained, the characteristic removal values are specified. For overlapping pulses of optimum fluence, the removal rate is as high as $27.8 \text{ mm}^3/\text{min}$ for aluminum, $21.4 \text{ mm}^3/\text{min}$ for copper, $15.3 \text{ mm}^3/\text{min}$ for stainless steel, and $129.1 \text{ mm}^3/\text{min}$ for Al_2O_3 , when a laser beam of 187 W average laser powers irradiates. On stainless steel, it is demonstrated that the removal rate increases to $23.3 \text{ mm}^3/\text{min}$ when the laser beam is very fast moving. This is thanks to the low pulse overlap as achieved with 800 m/s beam deflection speed; thus, laser beam shielding can be avoided even when irradiating high-repetitive 20-MHz pulses. © 2016 Society of Photo-Optical Instrumentation Engineers (SPIE) [DOI: 10.1117/1.OE.55.9.096109]

Keywords: ultrashort pulse laser; high-average power; high-rate; high-throughput; polygon; microprocessing.

Paper 160508P received Apr. 6, 2016; accepted for publication Sep. 1, 2016; published online Sep. 22, 2016.

1 Introduction

Ultrashort pulse lasers are attracting an increasing interest in micromachining thanks to their unique properties in terms of unprecedented precision and quality, high removal efficiency, and avoidance of thermal interaction during processing. The main disadvantage of the commercially available ultrashort pulse laser systems is the low average laser power that results in inadequate low processing speeds and throughput and is considered as main drawback of this promising technology for wide industrial applications. However, with the recent developments of multihundred Watt lasers,^{1–3} powerful systems are now on hand providing ultrashort pulses with tens of megahertz repetitions and sufficiently high energies for efficient ablation.

So far, only a few research studies have systematically investigated power scaling in metal ablation^{3–5} reporting a removal rate of up to $9.5 \text{ mm}^3/\text{min}$. This is ~ 20 times faster than conventional ultrashort pulse laser processing using low average laser power.⁵ In these studies, multi-MHz repetitive pulses were irradiated with inadequate low scan speeds that caused high thermal load to the substrate as a result of high pulse overlaps. As a result, material melting and artifacts, such as cone-like protrusions and microholes, originated at the laser-processed surfaces that lowered the machining quality and are unacceptable for most micromachining applications. However, the literature shows that laser processing using ultrashort pulses of optimum fluence and lateral pulse spacing in the range of the focus radius^{6,7} produces high-efficient material removal along with high processing quality. According to this, for efficient micromachining with multihundred Watt lasers, ultrashort pulses of microjoule energy

should be irradiated at tens of megahertz repetitions. In fact, this requires very high laser beam deflection speeds of hundreds of m/s to retain the optimum pulse spacing of a few micrometers at megahertz pulse repetitions. Such high speeds cannot be supplied by conventional (galvanometer) scan systems, and so, the development of alternative beam deflection methods is a current research challenge. Currently, there have been a few attempts reported using rotating cylinders, electro- and acousto-optical deflectors, or fast rotating polygons for high-speed beam deflection.^{8–11} In another approach, multifacet mirror (polygon)-based biaxial scan systems were engineered^{12,13} for two-dimensional (2-D) raster scanning with the aim to further enhance laser beam scanning speed and functionality for flexible microproduction.

In this paper, high-throughput laser machining is investigated using a high-speed polygon mirror-based scan system in conjunction with a picosecond laser supplying multihundreds Watt average laser powers to the substrates. The main goal of this study is to determine the optimum processing conditions for efficient material removal on different metals and Al_2O_3 ceramic substrates. For this, cavities are produced in the sample substrates by varying the fluence, pulse repetition frequency, average laser power, and scan speed. The depths of the made cavities are measured and analyzed in terms of achieved volume ablation rate, material removal rate, and removal efficiency. In addition, the machining quality is evaluated by scanning electron microscope (SEM) micrographs taken from the produced cavity bottom surfaces. Finally, based on the results obtained, the optimum processing parameters for efficient and high-quality high-throughput ultrashort pulse laser processing are presented.

*Address all correspondence to: Joerg Schille, E-mail: schille@hs-mittweida.de

2 Polygon Scan Technique

Since the last decade, the technical development of polygon mirror-based scan technique is a key research area at the Laserinstitut Hochschule Mittweida, driven by implementing high-average power lasers in high-speed micromachining processes with the aim to significantly increase throughput and productivity. When using a multifacet mirror or polygon, fast rotating around a mechanical axis at constant speed, each time the respective facet sweeps the incoming laser beam along one axis at very high speed. By focusing this fast moving laser beam through an f -theta objective to a sample surface, a number of fast recurring lines are produced, deflecting the laser beam in one direction across the substrate surface. Each scanned line consists of hundreds of resolvable spots that can be electronically addressed pixelwise. For biaxial high-speed processing, the laser beam is raster-scanned across the substrate by relatively moving the beam in the second dimension when it flips from one facet to the other. This second dimension deflection of the laser beam is essential for 2-D processing and has been done so far by synchronizing the laser and a polygon line scanner with a linear stage, which moves the sample perpendicularly to the polygon scan direction.^{4,10}

As an alternative, the Laserinstitut Hochschule Mittweida has been engineering and testing different concepts using polygon mirrors, for fast beam movement, in combination with a galvanometer scan unit to deflect the laser beam in the second direction, as schematically shown in Fig. 1. In this figure, the given fast axis and slow axis indicate the deflection of the laser beam either by the polygon or the galvanometer scanner, respectively. To separate the individual lines scanned with the respective facets and further to avoid undefined irradiation states, the beam path is shut-off at the time when the laser beam passes the edges between two facets.

The difference between the architecture of scanner “concept I and II” is that the galvanometer scan unit is mounted either in front [Fig. 1(a)] or behind [Fig. 1(b)] the rotating

polygon mirror wheel. In concept I, to deflect the laser beam in the second dimension (slow axis), the beam is moving along the facet widths, and thus, a wider width of the polygon mirror wheel is required. As a consequence, the polygon mass is comparatively high, which in turn lowers both maximum achievable rotation speed and dynamic of the polygon. Another disadvantage of concept I is the high spatial distortion of the scanning field, pointed out in Fig. 1(d). The input data in this theoretical estimate, however, implied uniform and equidistant point distribution within the scan field. It is noteworthy that this theoretical scan field analysis was performed without considering the f -theta focusing lens. In consequence, the identified distortions are not caused by optical aberrations but are rather the result of the varying position of the laser beam when it reflects on the polygon facet.

This concept I scanner architecture was initially studied in high-speed microprocessing using a 3-kW laser beam, delivered by a high-brilliant continuous-wave fiber laser and remarkable scan speeds above 300 m/s that was presented for the first time.¹² Meanwhile, in our laboratories, unprecedented laser beam scan speeds above 2000 m/s have been demonstrated within a $700 \times 700\text{-mm}^2$ scanning field. This was achieved using the above concept I polygon scanner in combination with a 1235-mm long focal-length objective that focuses the laser beam in a minimum spot of $75\text{-}\mu\text{m}$ diameter.

In concept II, the laser beam is deflected by the galvanometer scanner after passing the polygon. In this configuration, the polygon mirror wheel can be designed much smaller, which enhances the performance of the polygon scan unit but, as a detrimental effect, the size of the galvanometer scan mirror paddle enlarges. The spatial distortion of the scanning field given in Fig. 1(e) shows a symmetric aberration in slow axis while point spacing increases in fast axis. This fast axis aberration results from the moving reflection point of the laser beam along the polygon facet, as already discussed above for concept I, and is corrected in “concept

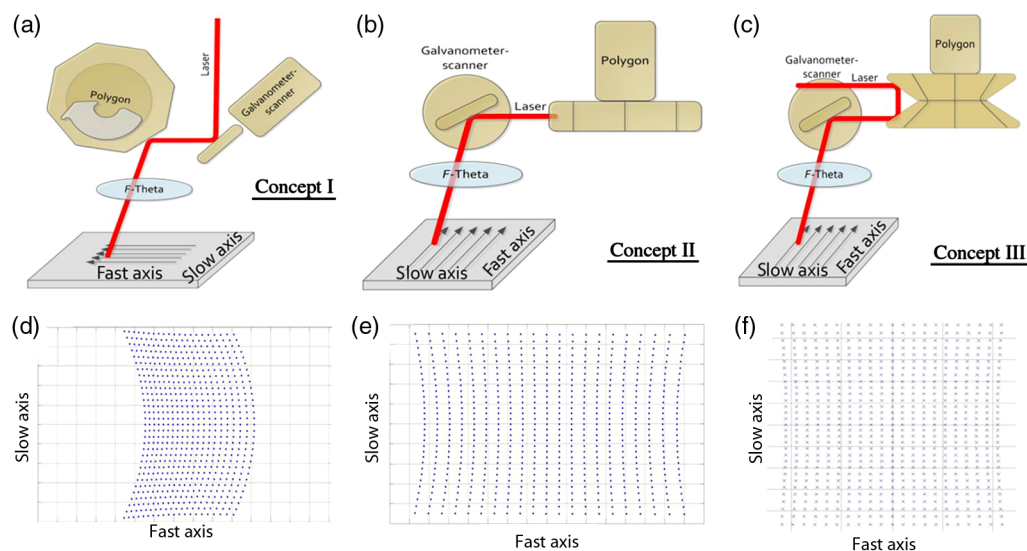


Fig. 1 Different polygon concepts engineered and tested at the Laserinstitut Hochschule Mittweida, the difference between the concepts is that the galvanometer scan unit, used for biaxial beam deflection, is mounted either (a) in front or (b) behind the polygon mirror wheel, while double-reflecting polygon architecture was used in (c); the respective scanning fields are presented in (d) to (f).

III” using a double-reflecting polygon mirror architecture schematically shown in Fig. 1(c). For the concept III architecture, Fig. 1(f) illustrates a scanning field that is mirror-symmetrically distorted in fast and slow axis. This distortion can be corrected electronically by standard f -theta lens correction procedures in slow axis and by using the in-house developed fast laser beam position control unit in fast axis. The hardware architecture of concept III is shown in Fig. 2(a).

However, there is a further incongruity between the expected and real laser beam positions that varies from facet to facet and is the result of limited precision when manufacturing the polygon mirror device. The effect of the pyramidal and tangential errors, both describing the deviations of the made polygon from an ideal geometric profile, is indicated in Fig. 2(b). In the upper figure, it becomes evident that the laser beam is differently shifted with each individual facet in consequence of the geometric polygon errors while the lower figure demonstrates exact ablation using electronically corrected laser beam positioning. The positioning accuracy of this concept III was better than $10\ \mu\text{m}$ that can be guaranteed for scan speeds up to $1000\ \text{m/s}$. The control software of the polygon scanner is featured with a vector engine for real-time slicing and process control that enables 2-D texturing and 2.5D engraving with very high laser beam scanning speeds, as demonstrated in Figs. 2(c) and 2(d) by machining examples.

3 Experimental Details

3.1 Experimental Setup

High-throughput laser machining was studied using the concept III polygon scanner in conjunction with a high-average power, high pulse repetition frequency picosecond laser system (PX 400, EdgeWave GmbH). The emitted laser beam was linearly polarized, the wavelength was $1064\ \text{nm}$, and the pulse width was $10\ \text{ps}$. The pulse repetition frequency was adjustable in the range between 1 and $20\ \text{MHz}$. The highest pulse energy was available for 1-MHz pulses at maximum laser power and reduced with increasing pulse repetition frequency. The maximum average laser power emitted at the laser beam exit port was $270\ \text{W}$, delivered at 20-MHz pulse repetition frequency, and decreased to $220\ \text{W}$ when pulses of lower repetition frequency were used.

The laser beam was enlarged by a beam expander and focused through a 420-mm f -theta lens to a minimum focus spot of $58\text{-}\mu\text{m}$ diameter at the processing plane. Using these optical configurations, the maximum beam scanning speed v_{max} of $1000\ \text{m/s}$ was reached at the highest polygon rotation speed. However, to estimate the achievable throughput and productivity, the effective processing speed v_{eff} has to be considered, which is influenced by the maximum utilization rate of the polygon facet $\eta_{f,\text{max}}$ to be

$$v_{\text{eff}} = \eta_{f,\text{max}} \cdot v_{\text{max}} = \frac{l_{\text{eff}}}{l_{\text{max}}} \cdot v_{\text{max}}. \quad (1)$$

The utilization rate is given by the ratio between the theoretical scan length l_{max} and the effective processable scan length l_{eff} , which is limited by the technical architecture of the polygon and the used optical components. This can be seen in Fig. 3(a) showing the maximal achievable scan length for an infinite small laser beam moving from one facet edge to the other through an endless aperture to the sample. In practice, as illustrated in Fig. 3(b), the laser beam starts and stops moving along the facets with a distinct displacement to the facet edges that are determined by the size of laser beam at this position. Further, the effective reachable processing length is limited by the finite aperture size of the focusing lens. For the concept III polygon scanner, the maximum and effective scanning lengths are measured to be 668 and $325\ \text{mm}$, yielding a utilization rate of 49% . According to Eq. (1), the effective processing speed of the polygon scanner architecture concept III is $490\ \text{m/s}$ and will be reached when the entire length of the effective processable length is processed at maximum 1000-m/s scan speed. A summary of the constant and varied processing parameters investigated in this study is presented in Table 1.

3.2 Processing Strategy and Analysis Methods

In this series of experiments, rectangular cavities were produced in industrial-grade stainless steel (X5 CrNi 18-10, AISI 304), aluminum [EN AW-Al Mg1(B)], copper (99% purity), and $96\% \text{Al}_2\text{O}_3$ ceramic plates by repeated scanning the laser beam line-by-line and layer-by-layer across the substrate surface. The processing parameter settings were

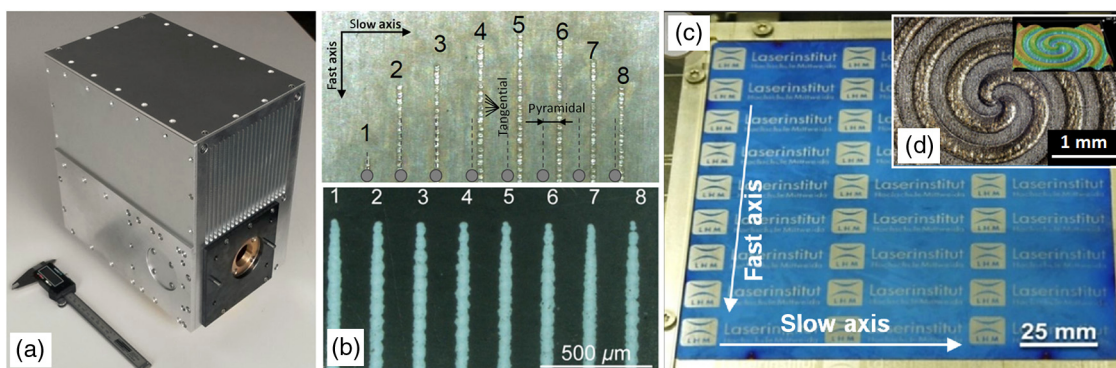


Fig. 2 (a) Concept III polygon scan unit; (b) top: laser beam starting from offset positions as identified for each individual polygon facet and bottom: the start position is electronically corrected; (c) institute logo processed on SiN-layered silicon wafer using $200\ \text{m/s}$ scan speed, $840\ \text{cm}^2/\text{min}$ processing rate was achieved with 400-W continuous wave laser power; (d) 2.5D-engraving structure produced in stainless steel using 400-W continuous wave laser powers and 200-m/s scan speed.

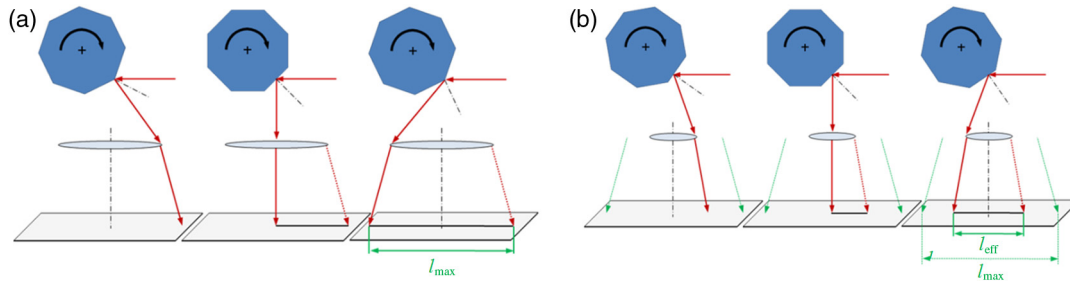


Fig. 3 (a) Maximum achievable scan length for an infinite small laser beam moving from one facet edge to the other through an endless aperture and (b) start and end position of the laser beam on the facets, the displacement to the edges depends on both laser beam spot size and free aperture through the system.

Table 1 Summary of constant and varied processing parameters.

Constant parameter settings	
Parameter	Value
Wavelength λ	1064 nm
Pulse duration τ_H	10 ps
Beam quality M^2	1.43
Focal length f	420 mm
Focus diameter $2w_0$	58 μm
Maximum facet utilization rate $\eta_{f,\max}$	49%
Studied parameter settings	
Parameter	Value
Average laser power P_{av}	18 to 187 W
Pulse repetition frequency f_R	1.01 to 20.3 MHz
Pulse energy Q_P	3.2 to 180 μJ
Peak fluence H_0	0.3 to 13.8 J/cm ²
Polygon scan speed $v_{\max}$	10 to 800 m/s

varied over a wide range while the irradiated average laser power increased with either higher pulse energy or higher pulse repetition frequency or both. The highest average laser power studied in this work was 187 W. This was the maximum laser power supplied by pulses of 1 MHz to the focal plane.

The depths of the made cavities were analyzed using a confocal point sensor (CF-4, nanoFocus GmbH) capable to measure surface topographies with submicrometer resolution. From the measured cavity depth d_C , which is influenced by the laser processing parameter settings, the volume ablation rate V_{SP} defining the ablated volume per irradiated laser pulse is determined as follows:

$$V_{SP} = \frac{d_P \cdot d_H \cdot d_C}{n_S}, \quad (2)$$

where d_P is the pulse-to-pulse spacing in scan direction, d_H is the hatch distance between the scanned lines, and n_S is the

number of scan crossings. According to Eq. (3), the material removal rate $\text{MRR}_{\max}$, which is the achievable maximum removable volume per time interval, is calculated by scaling the volume ablation rate obtained with the pulse repetition frequency f_R

$$\text{MRR}_{\max} = V_{SP} \cdot f_R. \quad (3)$$

In practice, with regards to the productivity, the effective scan speed has to be taken into account, so the effectively achievable processing rate MRR_{eff} calculates by the product of maximum achievable material removal rate and utilization rate of the polygon scan system to be

$$\text{MRR}_{\text{eff}} = \text{MRR} \cdot \eta_{f,\max}. \quad (4)$$

In order to identify the optimum parameters for efficient material removal, the removal efficiency is evaluated by relating the ablated volume per pulse to the irradiated pulse energy, or equivalent, the material removal rate to the irradiated average laser power P_{av} is given as

$$\text{Removal efficiency} = \frac{V_{SP}}{Q_P} = \frac{\text{MRR}_{\max}}{P_{\text{av}}}. \quad (5)$$

In addition to the depth measurements, for quality analysis, the bottom surfaces of the made cavities are examined in this study by micrographs taken with the JEOL JSM-6510 SEM.

4 Results and Discussion

Figure 4 presents the ablation results obtained from the analysis of the depths of cavities made in aluminum, copper, and stainless steel under varied processing parameter settings. The cavities were produced with constant hatch and pulse spacing of 10 μm . This was achieved by adjusting the scan speed to the studied pulse repetition frequency, i. e., 10 m/s at 1 MHz, 100 m/s at 10 MHz, and 200 m/s at 20 MHz, respectively. The graphs arranged on the left in Fig. 4 plot the volume ablation rate and the removal efficiency as a function of fluence, the pulse repetition frequency varies in the range between 1 and 20 MHz. It is shown for the three investigated metals, the volume ablation rate increases with increasing fluence while the highest volume ablation rate is only achieved with pulses of 1-MHz pulse repetition frequency. This is because the pulse energy, or rather fluence, was highest when pulses of 1-MHz pulse repetition frequency were irradiated at the maximum average laser power of 187 W investigated in this study. For pulses of

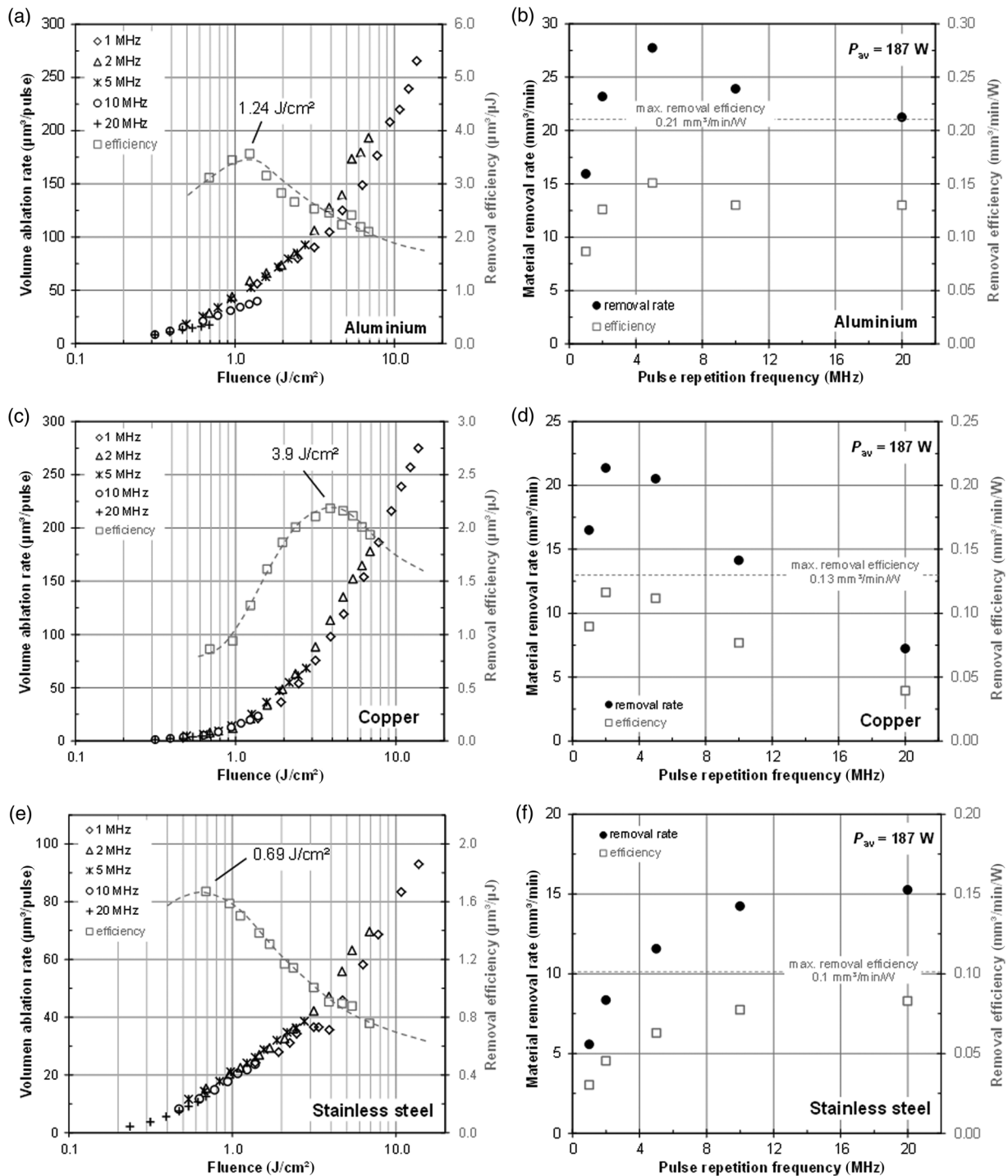


Fig. 4 Volume ablation rate versus fluence (left-hand) and material removal rate versus pulse repetition frequency (right-hand) as obtained on aluminum (a and b), copper (c and d), and stainless steel (e and f); in addition, the removal efficiency is presented.

higher repetition frequencies, however, the pulse energy decreased as a result of the linear relation between average laser power, pulse energy, and repetition rate. As a consequence, the maximum achieved volume ablation per pulse decreased with increasing pulse repetition rate that was due to the lower available pulse energy (or fluence). In addition, the graphs depict that metal removal is not significantly influenced by the pulse repetition frequency. This can be seen for pulses of identical fluence but different repetition

rates because almost the same amount of material was removed regardless of whether the pulse repetition frequency was 1, 5, or 20 MHz.

By comparing the removal efficiency, the left-hand graphs of Fig. 4 reveal a clear difference for the various metals. For stainless steel, the highest removal efficiency was found to be $1.67 \mu\text{m}^3/\mu\text{J}$, achieved with pulses of $0.69 \text{ J}/\text{cm}^2$ fluence. The highest removal efficiency detected on aluminum was $3.57 \mu\text{m}^3/\mu\text{J}$ at $1.24 \text{ J}/\text{cm}^2$ and for copper $2.18 \mu\text{m}^3/\mu\text{J}$

Table 2 Summary of the identified optimum processing conditions for high-throughput machining of metals and Al_2O_3 ceramic, the given characteristic removal values are derived from the results presented above in Figs. 4 and 5.

	Aluminum	Copper	Stainless Steel	Al_2O_3
Ablation threshold H_0	0.16 J/cm ² ¹⁵	0.28 J/cm ² ¹⁵	0.09 J/cm ² ¹⁵	0.81...3.6 J/cm ² ^{16,17}
Optimum fluence H_{opt}	1.18 J/cm ²	2.07 J/cm ²	0.67 J/cm ²	10.8 J/cm ²
Maximum achieved removal efficiency	3.57 $\mu\text{m}^3/\mu\text{J}$ at 1.24 J/cm ² , 33.0 W, and 2 MHz	2.18 $\mu\text{m}^3/\mu\text{J}$ at 3.90 J/cm ² , 104 W, and 2 MHz	1.67 $\mu\text{m}^3/\mu\text{J}$ at 0.69 J/cm ² , 18.4 W, and 2 MHz	11.9 $\mu\text{m}^3/\mu\text{J}$ at 10.8 J/cm ² , 144 W, and 1 MHz
Maximum achieved material removal rate MRR_{max}	27.8 mm ³ /min at 187 W, 5.1 MHz	21.4 mm ³ /min at 187 W, 2.0 MHz	15.3 mm ³ /min at 187 W, 20.3 MHz	129.1 mm ³ /min at 187 W, 1.0 MHz
Effective processing rate MRR_{eff}	13.6 mm ³ /min at 187 W, 5.1 MHz	10.5 mm ³ /min at 187 W, 2.0 MHz	7.5 mm ³ /min at 187 W, 20.3 MHz	63.2 mm ³ /min at 187 W, 1.0 MHz

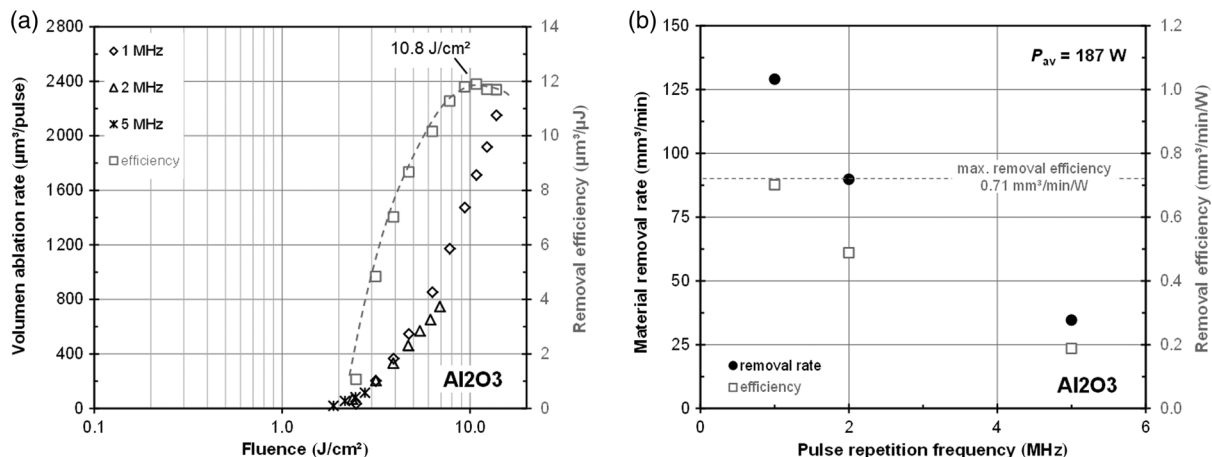
at 3.90 J/cm², respectively. It should be stated here that these experimentally identified efficient fluencies are in excellent agreement with the theoretical optimum fluence for ablation, considered to be 7.4 times above the ablation threshold.¹⁴ This is, in particular, for aluminum and stainless steel, where the optimum fluencies for megahertz pulse repetitions have been theoretically determined of 1.18 and 0.67 J/cm², as can be seen in Table 2. In the case of copper, by contrast, the theoretical optimum fluence of 2.07 J/cm² is about half as much as 3.90 J/cm² identified in this work for most efficient ablation, as shown in Fig. 4(c).

From an application point of view, the maximum material removal rate can only be achieved if pulses of most-efficient energy irradiate at maximum laser power, which, in turn, determines the operating pulse repetition frequency. This can be seen in the right graphs of Fig. 4, plotting the material removal rate achieved with 187-W average laser powers as a function of the pulse repetition frequency. For the studied metals, there is a rising tendency of the material removal rate with increasing pulse repetition frequency to maximum value, although the curves decrease with further progression, as can be seen for aluminum and copper.

For aluminum, the highest removal rate was 27.8 mm³/min, obtained with pulses of 5-MHz repetition

frequency and 2.76-J/cm² fluence. It is noteworthy that this fluence is twice the fluence of 10-MHz pulses of 1.38 J/cm² that is little above the optimum for material ablation as estimated of 1.18 J/cm². In consequence, it can be assumed that the lower removal rate, which was obtained for pulses of almost optimum fluence and higher repetition frequency, was caused by extended impact of particle shielding, taking place when pulses of shorter time intervals irradiate. This finding is supported by the analysis of the removal efficiency. In fact, the highest experimentally determined removal efficiency on aluminum was 3.57 $\mu\text{m}^3/\mu\text{J}$. By scaling this with the repetition rate, the maximum achievable removal efficiency is calculated to be 0.21 mm³/min/W, resulting in 38.6 mm³/min material removal rate for the 187 W applied laser powers. Figure 4(b) shows, however, that this high material removal was not reached in the experiments.

On the one hand, this is because the pulses were either above or below the optimum fluence for efficient ablation on aluminum since not all combinations of fluence and respective pulse repetition frequencies were investigated. On the other hand, shielding effects reduce the impinging optical energy when pulses of megahertz repetition frequency irradiate, decreasing the removal efficiency too.

**Fig. 5** (a) Volume ablation rate versus fluence and (b) material removal rate versus pulse repetition frequency as obtained for Al_2O_3 ceramic, in addition, the removal efficiency is presented.

As shown in Fig. 4(d), the maximum material removal rate achieved on copper is $21.4 \text{ mm}^3/\text{min}$, processed with 6.9 J/cm^2 and 2 MHz . This figure further indicates that $0.12 \text{ mm}^3/\text{W}/\text{min}$ as the highest removal efficiency achieved with 187-W laser powers is little below the estimated maximum removal efficiency of $0.13 \text{ mm}^3/\text{W}/\text{min}$. This is potentially due to the fact that the experimentally identified most efficient fluence of 3.90 J/cm^2 was not studied at maximum laser power. If doing so, the optimum pulse repetition frequency is 3.5 MHz that could be used at 187-W average laser powers. For this pulse repetition frequency, the removal rate calculates to be $24.1 \text{ mm}^3/\text{min}$, presupposing that the removal process is not affected by particle shielding. By assuming constant laser power settings, copper processing with pulses of higher repetition frequencies will lead to lower material removal rates, such as identified in Fig. 4(d) for the studied 187 W. This might be affected by particle

shielding, as mentioned above for aluminum, but as another effect, the fluence of the irradiated pulses was below the optimum.

For stainless steel irradiated with 187-W laser powers, the corresponding graph in Fig. 4(f) shows the increase of the removal rate to a maximum value of $15.3 \text{ mm}^3/\text{min}$. This was achieved with pulses of optimum fluence at 20-MHz pulse repetition frequency. However, if using this high pulse repetition frequency, shielding effects have detrimental influence on material removal. Since the removal efficiency is $0.1 \text{ mm}^3/\text{min}/\text{W}$ for pulses of optimum fluence and 2 MHz , as estimated from the results given in Fig. 4(e), it decreased to be $0.08 \text{ mm}^3/\text{min}/\text{W}$ when irradiating pulses of same fluence but the higher 20-MHz pulse repetition frequency, as can be seen in Fig. 4(f).

Figure 5 presents the characteristic removal values obtained for Al_2O_3 ceramic as a function of fluence and

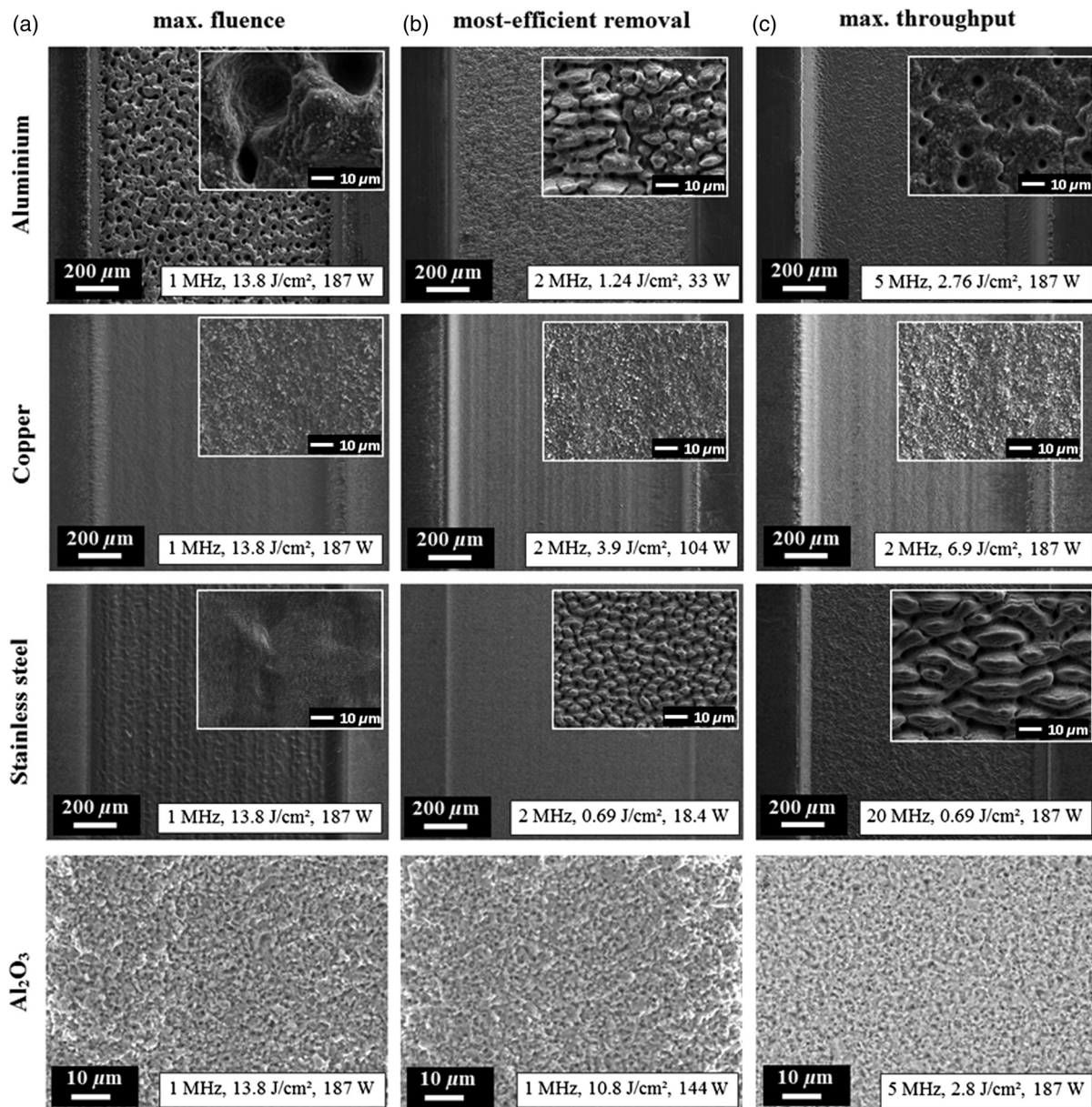


Fig. 6 SEM micrographs taken from the bottom surface of the cavities made by varying the processing parameter settings; (a) pulses of maximum fluence were applied; (b) parameter settings for most-efficient material removal were used; and (c) parameter settings for highest throughput were applied.

pulse repetition frequency. In Fig. 5(a), the highest volume ablation rate of $2152 \mu\text{m}^3$ per pulse is achieved with pulses of $13.8\text{-J}/\text{cm}^2$ fluence and 1-MHz repetition frequency, decreasing rapidly with faster pulse repetitions respective lower fluence. The maximum removal efficiency was $11.9 \mu\text{m}^3/\mu\text{J}$, obtained with pulses of $10.8\text{-J}/\text{cm}^2$ fluence. As a result of this high optimum fluence, the maximum material removal rate of $129 \text{ mm}^3/\text{min}$ was achieved when 187 W average laser powers irradiated at 1 MHz, as depicted in Fig. 5(b). It is also shown for this processing parameter set that the removal efficiency is almost equal to the maximum removal efficiency of $0.71 \text{ mm}^3/\text{min}/\text{W}$. This indicates for 1-MHz pulse repetition frequency there is less impact of shielding on material removal.

A summary of the identified optimum processing conditions for high-throughput laser machining of metals and Al_2O_3 ceramic material is given in Table 2. For the diverse materials, the ablation threshold and the optimum fluence (theoretical value) for efficient removal are included in the table. The ablation thresholds are referenced to the literature published elsewhere while the optimum fluence is estimated to be 7.4 times above the ablation threshold.¹⁴ In addition to the removal efficiency and the material removal rate, the effective processing rate is presented in order to estimate achievable throughput. Therefore, the maximal achieved material removal rate was matched with the utilization rate of the polygon scanner following Eq. (4).

In Fig. 6, SEM micrographs of the produced cavity bottom surfaces provide first impressions about the achieved processing quality when using high-average power lasers in micromachining. For each material, the cavity bottom surfaces shown on the left were made by applying pulses of $13.8 \text{ J}/\text{cm}^2$ maximum fluence available at 1 MHz and 187 W. Parameter settings for most-efficient removal or rather maximum throughput were applied to produce the cavities presented in the center and on the right, respectively. A substantial influence of the processing parameters on the surface quality can be seen. Thereby, the characteristic shape of the appearing (sub-) microsurface features and significant

melt formation varied depending on the irradiated laser power and fluence.

Remarkably, the surfaces produced at $13.8 \text{ J}/\text{cm}^2$ appeared very smooth that might be caused by remelting the surface as a result of the high amount of deposited thermal energy. However, the underlying smoothing mechanisms have not been fully clarified and further work is still needed. For pulses of lower fluence and higher pulse repetition frequency, the processed surfaces tend to be covered by rough and jagged microscaled features, as can be seen in particular on stainless steel and aluminum. This formation of cone-like protrusions has already been observed for megahertz femtosecond pulses and might be induced by heat accumulative effects.¹⁸ This has been proven in other works, i.e., reporting, that bumpy surface modification is caused by high surface temperatures during processing.¹⁹

A feasible method to decrease thermal loading and shielding in high-average power laser processing can be the increase of the lateral spacing between the impinging pulses. This is demonstrated in Fig. 7 on the example of stainless steel, where pulses of $0.71 \text{ J}/\text{cm}^2$ optimum fluence were irradiated at different pulse spacing. Therefore, the scan speed was scaled up with the pulse repetition frequency to very fast beam deflection speeds. On the one hand, it is shown in Fig. 7(a) that the volume ablation rate increased with larger pulse spacing that gives evidence of less shielding losses, even if pulses of higher repetition frequencies irradiated. For example, the removal rate increased to be $23.3 \text{ mm}^3/\text{min}$ with 187 W laser powers irradiated at 20 MHz pulse repetition frequency, 800 m/s scan speed, and corresponding $40 \mu\text{m}$ pulse spacing. This rate is about 25% larger than that achieved with identical laser powers and pulse repetition frequency but $10 \mu\text{m}$ pulse spacing. On the other hand, by comparing the results obtained with same pulse spacing and fluence but varied pulse repetition frequencies, as a general trend, the removal rate is higher for the faster pulse repetition frequencies. This can be attributed to heat accumulation that will have the potential to overbalance shielding losses. The increase of the removal

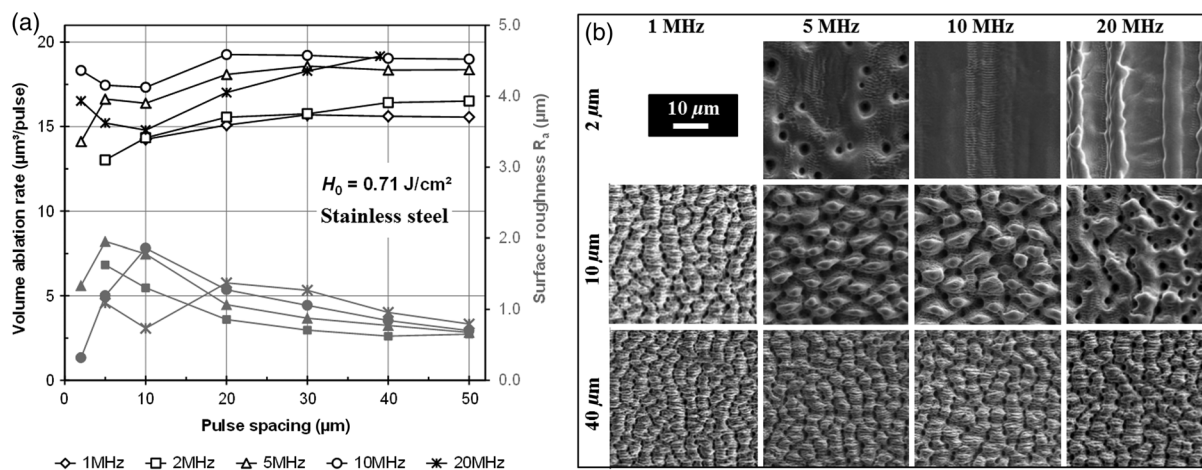


Fig. 7 (a) Volume ablation rate and cavity bottom surface roughness, achieved on stainless steel with pulses of $0.7 \text{ J}/\text{cm}^2$, as function of pulse spacing, the pulse repetition frequency was varied in the range between 1 and 20 MHz; (b) SEM micrographs of the cavity bottom surfaces produced by scaling the polygon scanning speed to very fast 800 m/s in order to vary both pulse spacing, ranging between 2 and $40 \mu\text{m}$, and pulse repetition frequency, ranging between 1 and 20 MHz that was achieved.

rate at higher pulse repetition frequencies can be seen for larger pulse spacing too, and as another advantage, the respective SEM micrographs in Fig. 7(b) shows relatively smooth cavity bottom surfaces.

The characteristic of the cavity surface bottoms made with larger pulse spacing appeared almost independent from the pulse repetition frequency and look very similar to the surface produced with 10- μm spacing at 1 MHz. This is also confirmed by surface roughness measurements given in Fig. 7(a), showing almost identical surface roughness R_a for pulses of 50- μm pulse spacing and varying pulse repetition frequencies. By contrast, cone-like protruded surfaces originated with faster pulse repetitions and small pulse spacing, while resolidified melt formations can be seen at the surfaces produced with both 10 and 20 MHz and 2- μm pulse spacing. The material melting observed seems to be induced by high thermal energy deposition because high amounts of laser energy irradiated with high pulse overlaps at very short time intervals at these parameter settings. However, as another effect, plasma absorption of the laser beam and plasma reheating cannot be ignored that increased thermal load to the substrates too.

These results indicate that ultrafast scanning of the laser beam is an option to avoid detrimental shielding and thermal loading effects when applying high-average power lasers in micromachining. It is demonstrated in this experimental study that polygon mirror-based scan techniques are feasible for high-efficient and melt-free high-throughput machining. However, exact synchronization between the ultrashort laser and the polygon scanner is still a key issue for future work. This is associated with the demand for fast switchable lasers, as needed for high-resolution position control of the laser beam within the scanning field. By solving these problems, high-average power ultrashort pulses lasers can be considered as key enabling technology with the potential to substitute standard manufacturing technologies in modern microfabrication.

5 Summary

High-throughput machining of industrial grade metals and Al_2O_3 ceramic was studied by employing a high pulse repetition frequency picosecond laser in conjunction with a unique, in-house developed biaxial polygon mirror-based scan system. At the beginning, different polygon concepts were engineered and tested with the aim to find out the best architecture for high-speed laser beam scanning usable in high-productive and high-precision microproduction.

In this study of high-throughput ultrashort pulse laser machining, removal rates as high as 27.8 mm^3/min for aluminum, 21.4 mm^3/min for copper, 15.3 mm^3/min for stainless steel, and 129.1 mm^3/min for Al_2O_3 ceramic were achieved when a 187-W average laser power irradiated at optimum pulse repetition frequency. For small 10- μm lateral pulse spacing, it was shown that beam shielding induced by plasma/particle interaction but also thermal effects have detrimental impact on material removal and machining quality. This is confirmed by SEM-micrographs showing molten or rather rough and cone-like protruded bottom surfaces of the cavities produced that resulted from high thermal load of the substrates if processing with high-average laser power. Significant larger pulse spacing, by contrast, produced melt-free surfaces of minimum roughness. This is

demonstrated by the example of stainless steel, where the lateral pulse spacing was varied between 2 and 50 μm by adjusting the polygon scanning speed to the pulse repetition frequency. By doing so, the removal rate increased to be 23.3 mm^3/min , obtained with 187-W average laser powers, 20-MHz pulse repetition frequency, 0.71-J/ cm^2 optimum fluence, and 40- μm pulse spacing. In conclusion, this experimental study demonstrates that synchronized high-speed beam deflection by using polygon scanners can overcome shielding and thermal loading as major limitations of multi-MHz ultrashort pulses. As a result of the low pulse overlap that was made possible to the very fast scan speeds, high-average power laser systems can be used very effectively in high-throughput machining, which is a major achievement.

Acknowledgments

The presented results have been conducted in course of the projects “Innoprofile Transfer–Rapid Micro/Hochrate-Laserbearbeitung” (No. 03IPT506X) and GROMinaS–Großflächige Mikro-/Nano-Strukturierung (No. 03FH037 PX4) funded by the Federal Ministry of Education and Research.

References

1. T. Eidam et al., “Femtosecond fiber CPA system emitting 830 W average output power,” *Opt. Lett.* **35**, 94–96 (2010).
2. P. Russbuehler et al., “Compact diode-pumped 1.1 kW Yb:YAG Innoslab femtosecond amplifier,” *Opt. Lett.* **35**, 4169–4171 (2010).
3. K. Du, S. Bruening, and A. Gillner, “High-power picosecond laser with 400 W average power for large scale applications,” *Proc. SPIE* **8244**, 82440P (2012).
4. B. Jaeggi et al., “High throughput ps-laser micro machining with a synchronized polygon line scanner,” *Proc. 8th Int. Conf. Photonic Technologies LANE*, 1–8 (2014).
5. J. Finger, C. Kalupka, and M. Reininghaus, “High power ultra-short pulse laser ablation of IN718 using high repetition rates,” *J. Mater. Process. Technol.* **226**, 221–227 (2015).
6. J. Schille et al., “Micro processing of metals using a high repetition rate femto second laser: from laser process parameter study to machining examples,” in *Proc. of ICALEO*, paper M102, pp. 1–10 (2011).
7. J. Schille et al., “Process optimization in high-average-power ultrashort pulse laser microfabrication: how laser process parameters influence efficiency, throughput and quality,” *Appl. Phys. A* **120**(3), 847–855 (2015).
8. S. Zeng et al., “Femtosecond pulse laser scanning using acousto-optic deflector,” *Sci. China Ser. G* **52**(5), 685–692 (2009).
9. S. Bruening et al., “Ultrafast scan techniques for 3D- μm structuring of metal surfaces with high repetitive ps-laser pulses,” *Phys. Proc.* **12**, 105–115 (2011).
10. R. De Loor, “Polygon scanner system for ultra short pulsed laser micro-machining applications,” *Phys. Proc.* **41**, 544–551 (2013).
11. G. R. B. E. Roemer and P. Bechtold, “Electro-optic and acousto-optic laser beam scanners,” *Phys. Proc.* **56**, 29–39 (2014).
12. H. Exner et al., “High rate laser micro processing using high brilliant CW laser radiation,” in *Proc. of the 11th Int. Symp. on Laser Precision Microfabrication*, Stuttgart, Germany, pp. 1–6 (2010).
13. A. Streek and S. Kloetzer, “Polygonscanteknik für die Lasermikrobearbeitung,” *J. Univ. Appl. Sci. Mittweide* **4**, 82–85, (written in German), in Scientific Reports (2015).
14. G. Račiukaitis et al., “Use of high repetition rate and high power lasers in microfabrication: how to keep the efficiency high?” *J. Laser Micro/Nanoeng.* **4**(3), 186–191 (2009).
15. J. Schille, “Investigation of micromachining using a high repetition rate femtosecond fibre laser,” Thesis, The University of Manchester, Manchester, UK (2013).
16. U. Engelhardt, J. Hildenhausen, and K. Dickmann, “Micromachining using high-power picosecond lasers,” *Laser Tech. J.* **8**(5), 32–35 (2011).
17. D. Ashkenasi, R. Stoian, and A. Rosenfeld, “Single and multiple ultra-short laser pulse ablation threshold of Al_2O_3 (corundum) at different etch phases,” *Appl. Surf. Sci.* **154**, 40–46 (2000).
18. J. Schille et al., “Micro structuring with highly repetitive ultra short laser pulses,” in *Proc. of the 9th Int. Symp. on Laser Precision Microfabrication (LPM)*, Quebec, Canada (2008).
19. F. Bauer et al., “Heat accumulation in ultra-short pulsed scanning laser ablation of metals,” *Opt. Express* **23**, 1035–1043 (2015).

Joerg Schille graduated in physical engineering from the University of Applied Sciences Mittweida in 2003, and received his PhD from The University of Manchester in 2013. The PhD study was centred around interaction phenomena using high-pulse repetition frequency ultrashort pulse lasers in micro machining. He is currently a member of the Rapid Micro Tooling group at the Laserinstitut Hochschule Mittweida. His research activities include laser micro processing, ultrashort pulse laser ablation, and high-rate laser processing.

Lutz Schneider graduated in physical engineering from the University of Applied Sciences Mittweida in 2009. Since then he is working as a R&D engineer at the Laserinstitut Hochschule Mittweida. He is an expert in ultrashort pulse laser ablation, laser micro machining, and high-rate laser processing. Actually he investigates laser technologies for large area surface processing and micro- and nanoscale surface texturing.

André Streek was educated and graduated in physical engineering at University of Applied Sciences Mittweida in 2004. Till 2010 he worked as a research and development engineer in various fields of laser research. His profession is in additive manufacturing, high-rate laser processing and in development of high speed polygon scanning

systems. Since 2010 he worked additionally on his thesis of laser micro sintering. He holds various patents in the field of rapid prototyping and polygon scanning systems.

Sascha Kloetzer graduated in ceramic engineering from the Ingenieurschule fuer Elektrotechnik und Keramik Hermsdorf in 1994, and mechanical engineering from the University of Applied Sciences Mittweida in 1996. Since 1998, he is working as a research and development engineer at the Laserinstitut Hochschule Mittweida. He is a member of the Rapid Micro Tooling group and develops mechanical and electronic devices for laser micro and macro processing, additive manufacturing and high-rate laser processing including polygon scanners.

Udo Loeschner graduated in physical engineering from the University of Applied Sciences Mittweida in 1998. Thereafter he was a R&D engineer at the Laserinstitut Hochschule Mittweida and joined the Rapid Micro Tooling group in 2006 to investigate laser micro processing with short and ultrashort pulse laser technologies. He completed his PhD thesis at the Technische Universitaet Ilmenau in 2007. Since 2011, he is as a professor for laser production technology including high-rate laser processing.